



Digital Twin Benefits From Collaboration With IoT

Digital Twin is an upcoming technology that is considered to be the next big innovation. However, there is more to Digital Twin than simply providing simulation

Arnab Ghosh

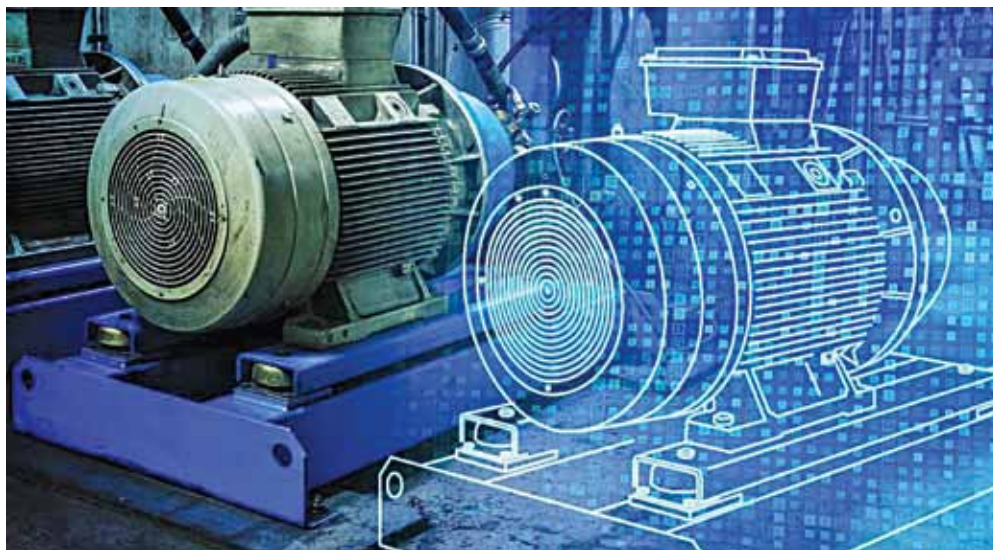
When we hear the word ‘twin,’ the image of a human twin possessing identical features such as the same eyes, lips, and nose comes to our mind. Digital Twin is similar to that. It is defined as the virtual representation of a physical object or system across its entire lifecycle, which uses real-time data or data from other sources to enable learning, reasoning, and dynamic recalibration for improved decision making.

As Digital Twin is constantly evolving, its definition varies from company to company. For instance, GE states that Digital Twin is a software representation of a physical asset, system, or process designed to detect, prevent, predict, and optimise through real-time analytics to deliver business value. In simple terms, Digital Twin is defined as a virtual model of a physical

thing. That thing could be a car, tunnel, bridge, jet engine, or anything else attached with multiple sensors whose function is to carry out data connection, which can be mapped to the virtual model.

But is Digital Twin all about simulation? Are physical world scenarios replicated in the digital world? Not exactly. Digital Twin initially starts as a simulation but goes on to perform real-time updates, unlike digital simulation. Results can be obtained by running, testing, and conducting assessments on a simulated version of a physical asset. Simulation is static and so does not know how the actual thing will behave in the real world. Parameters always need to be fed in the simulator to make it more realistic.

In short, a Digital Twin is the virtual representation of a system across its entire lifecycle. This includes its designing, operation, and production.



3D model of a pump, which is the twin of the original (Credit: Noria Corporation)

Early use of Digital Twin

In the year 1970, NASA successfully launched the spacecraft Apollo 13 into space. However, due to some technical issues, it could not complete its intended mission and had to return to Earth. It is said that NASA had used the concept of Digital Twin back then at a time when IoT did not even exist.

The story goes like this: Before the space mission began, a simulator was built to train the